

Mengqing Hu

(+49) 1623382989 | mengqinghu688@gmail.com | linkedin.com/in/mengqinghu | github.com/mengqing-hu

AI Engineer — Full-Stack Engineer

EDUCATION

TU Dresden

M.Sc. Computational Modeling and Simulation (Visual Computing)

Henan University of Technology

B.Sc. Software Engineering

Dresden, Germany

2023.10 – 2026.11 (Expected)

Zhengzhou, China

2019.09 – 2023.07

WORK EXPERIENCE

Helmholtz-Zentrum Dresden-Rossendorf (HZDR)

Computer Vision Intern

Dresden, Germany

2026.01 – Present

- Conduct a comprehensive survey of mainstream biomedical instance segmentation algorithms such as CPN, U-Net and Mask R-CNN, gain an in-depth understanding of CPN's contour regression mechanism based on Fourier descriptors, and conduct a comparative analysis of its technical characteristics and applicable scenarios relative to mask-based methods.
- Adapt the CpnU22 model to HZDR's public microscopy image dataset using the celldetection framework, execute percentile normalization, Albumentations-based data augmentation and dataset splitting to complete the entire data preprocessing pipeline.
- Tune hyperparameters including learning rate, batch size, NMS threshold and Fourier order, evaluate model performance by analyzing loss curves and average F1 scores across multiple IoU thresholds, and execute model training, inference and experimental validation on an HPC cluster.
- **Tech Stack:** Python, PyTorch, Computer Vision, Instance Segmentation, Contour Proposal Network

Fraunhofer IWU

Research Assistant

Dresden, Germany

2025.08 – 2026.03

- Developed an intelligent retrieval and QA system for research documents based on Large Language Models (LLMs), enabled batch parsing of PDF and Word files, text cleaning, chunking, and vectorization. Employed a chunking strategy optimized via semantic sentence splitting + overlapping sliding windows to mitigate semantic fragmentation issues in long documents, resulting in a 15% improvement in retrieval recall accuracy.
- Constructed a hybrid recall strategy combining BM25 sparse retrieval with dense vector retrieval to address the limitations of pure semantic retrieval, boosted the Top-5 hit rate from 72% to 91%. Integrated a re-ranking mechanism to filter out irrelevant snippets, thereby effectively enhancing the quality of the generated context.
- Utilized Prompt Engineering to constrain the model to generate responses solely based on the retrieved context, thereby mitigating the risk of hallucinations. Supported private deployment, achieved a 35% improvement in QA accuracy while maintaining an average response time of under 1 second, significantly boosting the efficiency of internal document retrieval.
- **Tech Stack:** Python, LangChain, RAG, BM25, Chroma, Prompt Engineering, Cross-Encoder Reranker

TU Dresden – Institute of Mechatronic Engineering

Research Assistant

Dresden, Germany

2024.11 – 2025.04

- Implemented sliding-window anomaly detection, mask generation, histogram matching, and block-based dataset partitioning for OCT tomographic images in additive manufacturing to achieve automated binary classification labeling (Good/Bad).
- Built a lightweight classification model based on ResNet26-V2, designed a BottleneckV2 residual structure, and optimized hyperparameters including network width coefficient K, number of modules N, Dropout rate, and learning rate.
- Performed model interpretability analysis using t-SNE and Grad-CAM; achieved a final binary classification accuracy of 94.34%, enabling real-time monitoring of internal defects and quality classification within the additive manufacturing process.
- **Tech Stack:** Python, PyTorch, Computer Vision, Supervised Learning, CNN, t-SNE

PROJECT EXPERIENCE

Multimodal Video Understanding and Automatic Summarization

Dresden, Germany

Master Thesis Project (ScaDS.AI Dresden/Leipzig)

2026.04 – Present

- Design and implement an end-to-end multimodal video analysis pipeline that automatically handles data acquisition, audio-visual extraction, and content parsing directly from video URLs; perform keyframe extraction, scene segmentation, and semantic partitioning to establish a structured data foundation for long-form video understanding.
- Integrate technologies such as Whisper for speech transcription, CLIP/BLIP for visual-semantic understanding, OCR for text recognition, and speaker diarization to conduct cross-modal joint analysis of audio-visual information, thereby constructing structured semantic representations of video content.
- Leverage Large Language Models (LLMs) to generate automatic summaries of video content; combine semantic partitioning with automatic chapterization to segment long videos into thematic clips and generate narrative-style summaries and commentaries, thereby enhancing the accessibility and comprehension efficiency of public video information.
- **Tech Stack:** Python, Whisper, CLIP, BLIP, GPT, OCR, Pyannote, Speaker Diarization, LLaMA

OCT Image Feature Extraction and Clustering Analysis

Dresden, Germany

Research Project (TU Dresden – Institute of Mechatronic Engineering)

2025.03 – 2025.07

- Leveraged a dataset of over 8,000 OCT B-scan images for research into online quality monitoring in additive manufacturing, designed a comprehensive preprocessing pipeline incorporating binarization, morphological enhancement, and spline-based ROI extraction to effectively suppress noise in industrial images and provide high-quality input data for subsequent modeling.
- Designed and implemented a convolutional autoencoder for unsupervised feature learning, compressed 224×224 images into a 128-dimensional latent feature space; fine-tuned key hyperparameters such as Dropout and learning rate, and achieved a Silhouette Coefficient of 0.9101 in structural separation tasks, significantly outperforming pre-trained CNN models such as ResNet50 and VGG16.
- Employed PCA dimensionality reduction and K-Means clustering to achieve structural pattern recognition, yielded a Calinski-Harabasz Index of approximately 9.2×10 and a Davies-Bouldin Index as low as 0.20; validated feature separability through t-SNE and KDE visualizations, providing an effective solution for unsupervised industrial defect detection.
- **Tech Stack:** Python, PyTorch, Computer Vision, Unsupervised Learning, PCA, K-Means, t-SNE

Enterprise Knowledge Retrieval and Information Extraction System

Dresden, Germany

Team Project (FSD FahrzeugSystemDaten GmbH)

2024.05 – 2024.09

- Developed the web interface for an enterprise AI assistant using the Django framework, implemented modules for user authentication, chat history management, document uploading, and knowledge-based QA; built over six core functional pages, supporting multi-role permissions and concurrent access for multiple users.
- Participated in the development of the RAG retrieval pipeline, assisted with the parsing of technical documentation, text chunking, and vectorization; collaborated with the team to construct Elasticsearch indices and optimize the retrieval chain, thereby boosting the Top-5 document hit rate to over 80%.
- Engaged in front-end/back-end interface debugging and system integration testing; integrated with Large Language Model (LLM) service APIs to ensure stable interaction between the web application and the QA system, supporting unified search services for tens of thousands of enterprise technical documents.
- **Tech Stack:** Python, Django, LangChain, Elasticsearch, LLM, RAG, LLaMA, Mistral

TECHNICAL SKILLS

AI & LLMs: RAG, Prompt Engineering, LangChain, LangGraph, PyTorch, Scikit-learn, Chroma, FAISS

Backend Development: Python, FastAPI, Django, PostgreSQL, MySQL, Redis

Frontend Development: TypeScript, React, Next.js

Engineering & Deployment: Git, Linux, Docker, CI/CD, Azure

LANGUAGES

Chinese (C2) — English (C1) — German (A2)